THE UPIC SYSTEM: ORIGINS AND INNOVATIONS

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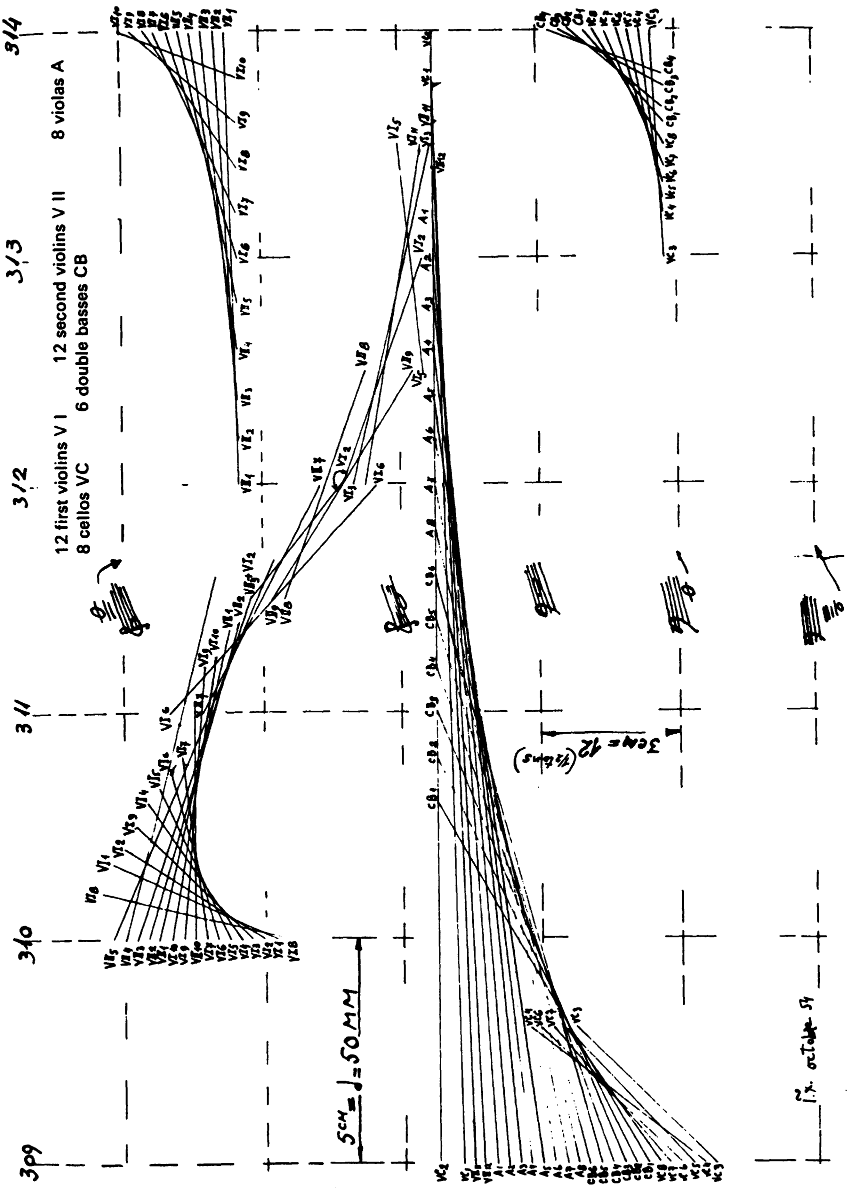
THE ORIGINS OF THE UPIC SYSTEM

THE IDEA OF THE UPIC system goes back to 1953–54, when Iannis Xenakis wrote music for orchestra, using graphic notation for representing musical effects that were too complicated to be specified with traditional staff notation. The work *Metastasis* (written in 1953–54 and premiered at Donaueschingen in 1955) makes systematic use of glissandi (continuous transition between two notes of different pitches). Xenakis drew the glissandi as straight lines in the pitch-versus-time domain. The score is written for sixty-one different instrumental parts. The great number of glissandi creates a sound space of continuous evolution comparable to the ruled surfaces and volumes that he used in architecture.

Writing the glissandi in sixty-one different orchestra parts by hand was quite arduous (see Example 1). Xenakis had then to transcribe the graphic



EXAMPLE 1: *Metastasis* (1954)

notation into traditional notation so that the music could be played by the orchestra. At this time, he came up with the idea of a computer system that would allow the composer to draw music. Indeed, graphic representation has the advantage of giving a simple description of complex phenomena like glissandi or arbitrary curves. Furthermore, it frees the composer from traditional notation that is not general enough for representing a great variety of sound phenomena. In addition, if such a system could play the score by itself, the obstacle of finding a conductor and performers who want to play unusual and "avant-garde" music would be avoided.

The idea that the composer could create autonomously was also essential. Also, in 1984, with the development of computer technology, real time became one of the first requirements. Indeed, a computer music system allows the composer to experiment with new musical ideas, listen to them, and modify them at will. If the system is fast enough, the composer gets the result of his work directly, so that the exchange between thought and ear is made very easy and immediate.

Another idea was to let the composer control and create all aspects of the composition: sound, symbols, syntax, and so forth. This means that the system should not impose predefined sounds, predefined compositional process, predefined structures, and so on. It is essential for the creative mind that ideas not go through theories or limitations that might not suit the composer.

It was also important that the composer could "work in a systematic way on various levels at the same time" (Lohner 1986), which is not the case in instrumental music or even in some computer music systems. This requirement implies that the hierarchy between sound design and musical architecture be absent or very flexible.

The UPIC system offers one solution to all these problems. The UPIC system has known several successive versions as computer technology has improved. The first version was born in 1977 and ran on a SOLAR mainframe. In 1983, the second version was based on two 8086 INTEL microprocessors. The first real-time version came in 1987, which increased the interaction between composer and machine tremendously. The newest PC/AT version (1991) that is described below is the first commercial UPIC system. It is totally real time and includes advanced "user-friendly" concepts in its graphic interface.

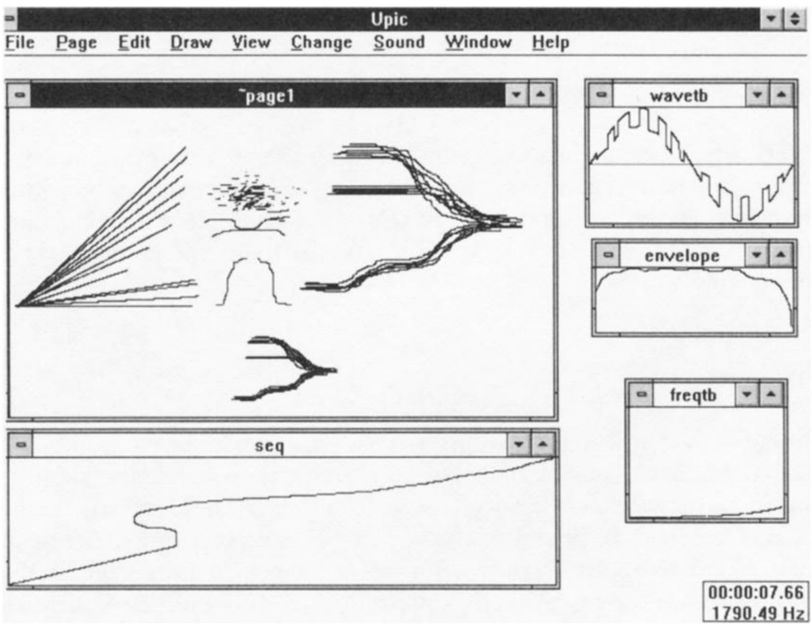
The UPIC Objects

The UPIC is a composition tool that offers the musician a notation based on a set of graphic objects. All of them are made of one or several graphs, depending on the type of object. Each object type has a specific function in the sound synthesis carried out by the machine. No other hidden object or parameter is used for this calculation, so that the composer has full control of the synthesis process.

Some object types, as the *envelope* or the *wave table*, are well known to the composer accustomed to electro-acoustics; other object types, such as the *page* or the *frequency table*, present an analogy with notions (score and musical scale) used in traditional notation, which make them easily understandable by all musicians.

Composing in the Pitch-Versus-Time Space: Working in the Page

The page is a set of pitch-versus-time graphs (see Example 2): time is represented from left to right, and pitch from bottom to top. Depending on the composition method, different meandings may be assigned to these graphs. The methods described below are given as simple examples of strategy. Many others are possible, and of course, they can be combined and applied within the same page.



EXAMPLE 2: THE UPIC OBJECTS

Traditional Method: The Page/Score Analogy

In a traditional score the notes represent sound events in a pitch-versus-time space. Notes are to be played by a specified instrument. Following the analogy of the page with a score, each of the graphs (named *arcs*) that compose the page is considered as a note, and two of the attributes of the arc (the wave table and the envelope) are supposed to describe the timbre of the instrument that plays the note. The UPIC provides tools to extract a wave table (a period of a signal) and an envelope (evolution over time of the amplitude of the sound) out of a sampled sound.

The Additive-Synthesis Method

The additive-synthesis method consists in superimposing the harmonics of the sound to be synthesized. In the page, each harmonic is represented by an arc (see above), whose associated wave table is one cycle of a sine wave. One may depart from theory by using any other wave table, including extracted ones. This illustrates the flexibility of the UPIC and its ability to combine different approaches.

The Frequency-Modulation Method

The frequency modulation of any group of arcs in a page requires two objects, an arc that represents the evolution over time of the modulating frequency, and an envelope that represents the evolution of the modulation index. A parameter specifies, for each modulated arc, the number of the modulating arc.

Pitch and Time

In traditional music, pitch and time (the two dimensions of the score) have a discontinuous structure, quite different from one civilization to another, and in each civilization from one style to another. These structures, the musical scale and the time scale, are important concepts in music theory, so they tend to be studied by themselves independently from any score. In the UPIC, too, the composer has the opportunity to work on these structures independently from the page. Afterwards, he will be able to test successively several of them with the same page.

In order to realize this independence, the UPIC uses two objects: the frequency table, which corresponds to the musical scale, and the sequence (which is also a table), which corresponds to the time scale.

THE FREQUENCY TABLE

The frequency table describes the musical scale and the frequency range of the page. It is a table used by the UPIC to convert the values of the vertical axis of the page (that represent pitches) into frequencies.

The range of frequencies of this table gives the range of frequencies accessible by the page. If the table is continuous, all the frequencies between the lower and upper limits of the table will be accessible. If the table has steps, the frequencies between the steps will not be accessible, like in a traditional musical scale. The UPIC provides tools to define tempered frequency tables where the octave is divided in equal steps (from 1 to 99). All types of musical scales, classical, microtonal, continuous, etc., and all the combinations between them, can be built with frequency tables.

THE SEQUENCE

The UPIC has two modes of playing a page. With the first mode, by analogy with the traditional way of playing a score, the page is played from left to right. At time zero, the read index (it points to the place on the page the UPIC has to play) is zero. In this mode, the tempo, defined by the composer, is constant. This means that every six milliseconds (the time resolution of the UPIC), a constant value (depending on the tempo) is added to the current read index.

In the second mode, every six milliseconds the UPIC enters the current time value in a table (called *sequence*), and reads the corresponding value. The value given by this table is the next read index, i.e. the next page place the UPIC has to play. This mode allows variations of tempo as well as reverse playing and jumps to noncontiguous places of the page.

Sound Design and Microstructure

The UPIC objects have a very fine resolution, for example: one hundredth of a tone for the frequency (the frequency table has 16K entries), six milliseconds for the sequence. This allows for very precise work on microstructures deep within the objects.

Let us take the example of the envelope: a graph that describes the evolution of sound amplitude (attack, decay, sustain, release)—it appears to be a very clear and simple notion, but what happens if we listen to the same sound with different envelopes, more and more complicated, that is, having more and more amplitude variations? Gradually one will no longer hear the amplitude variation, but only its frequency, that is, the envelope's effect is heard as pitch!

With a very fine scale, the borderline between the different UPIC objects becomes fuzzy, each notion encroaching on another's territory. Exploring the limits of these notions can result in quite unexpected and interesting effects.

Moreover, the great resolution of the UPIC objects to structure them internally in a very fine way. For example, it is possible to use a unique small curve as a pattern to draw an object, so as to give it an inner formal consistency.

Interactivity

Having shown the great importance assumed by graphic notation in the UPIC, it is time to relate these graphic forms to the musical form.

First of all, we have to avoid a possible misunderstanding: the UPIC is not a machine that would allow one to "hear" drawings. We insisted previously on the symbolic nature of the objects. Moreover, the eventual aesthetic quality of the drawing of an object does not guarantee an equivalent musical quality. Ultimately, since the UPIC is a music composition tool, music—not drawing in itself—should remain the focus.

Keeping in mind this perspective, the ability of the UPIC to execute immediately (in "real time") the commands of the composer takes on a great importance. In fact, when the composer's hand draws a UPIC object, it is guided by the eye, but also and above all, by the ear. The main advantage of "real-time" control is not the speed of execution but the autoregulation possibilities that it brings.

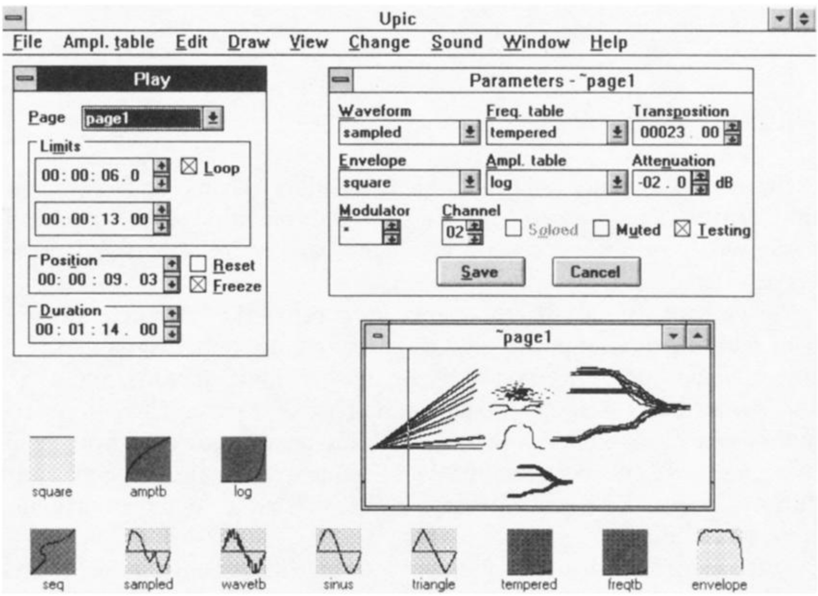
This method reintroduces some empiricism, and allows the use of aesthetic compositional criteria when modifying a graph or replacing an object with another.

Recording a Sequence

The composer may want to control in real time (as the UPIC allows him) the playing of a page, slowing down or accelerating the tempo, going backwards, or even jumping to another part of the page. This interpretation, with all its subtle variations of tempo, can be recorded as a *sequence* (see above for the definition of a sequence). The composer may use this recorded performance as it is or consider it a sketch and modify it by redrawing it.

Human Interface

The flexibility and originality of the UPIC rely on the conjunction of a "user-friendly" graphic interface and the power of real time. Indeed, the newest version of the UPIC is based on a standard PC/AT 386 microcomputer connected to a real-time synthesis unit. The PC handles the human interface which runs under Microsoft Windows 3.X. This powerful graphic interface presents standard pull-down menus and pop-up windows (see Example 3).



EXAMPLE 3: THE PAGE REAL-TIME CONTROLS

EDITING

The composer at the commands of the UPIC displays objects (waveform, envelope, page . . .) in resizable and zoomable windows. There is direct control over every part of the composition as all objects are available at the same time. A set of display facilities is provided: automatic arrangement of the objects in rows and columns, reduction into icons, overlapping of different kinds of objects one upon another.

The composer creates objects either by a command or by drawing. For example, a waveform can be extracted after sampling a natural sound or can be directly created by drawing. In any case, the user will be able to modify this waveform at will, simply by redrawing it: all the UPIC objects are redrawable.

Different drawing modes (freehand, broken lines, and so on) and the usual editing commands (cut, copy, paste) are available. Some other graphic transformations, such as symmetries and rotations, are also available. As an option for drawing, one can choose to employ a digitizer rather than a mouse, in order to reach the absolute coordinates directly. Furthermore, with a large digitizer, the whole work is available with total precision without the necessity to zoom in.

REAL-TIME CONTROLS

But the UPIC is not only a graphic score editor. Thanks to its computation capabilities, it allows for real-time interpretation of a score and, moreover, the real-time control of all the parameters of a sound, in its deepest details.

As we have seen above, the composer controls the interpretation of a page with the mouse: position on the page, tempo, tempo variations, etc. Instantaneous jumps from one page or another (four pages maximum) are also permitted as well as looping in any part of a page. The composer's motions are recorded in a sequence which can be redesigned later on. As all the objects are independent, the same sequence can also be applied to different pages. Two sequences are available (twelve minutes maximum each, six-millisecond accuracy).

Furthermore, in each page, as many as four groups of any number of arcs can be created with different types of selection (block select, list, or criteria). Every parameter of the selected arcs can be modified and the modification is instantaneously heard. For example, the frequency modulation effect generated by a modulating arc is very easily controlled by simply changing its average intensity; the lower the intensity, the lower the effect. A group of arcs can also be transposed, muted, or soloed and the temporary modifications can be saved.

Another impressive way to take advantage of real-time control is to draw or redraw an object directly. For example, when redrawing a waveform while a page is playing, one hears the global modification it imposes on the sound, including its interacitons with other timbres.

Performances

The real-time synthesis unit performs additive synthesis with frequency modulation on sixty-four simultaneous oscillators at 44.1 kHz (future extension to 128 oscillators).

Each oscillator has its own waveform, envelope, frequency table, amplitude table, frequency-modulating oscillator, and output channel. Sixty-four waveforms (4K entries), 128 envelopes (4K entries), four frequency tables (16K entries), four amplitude tables (16K entries), and sixteen output channels are available for any oscillator. Any oscillator can modulate any other, including itself. In order to avoid "clicks," envelopes are interpolated at each sample. Frequency variation is also interpolated. Digital audio interface is provided (AES/EBU and S/PDIF) for input and output at 44.1 kHz, 48 kHz, and 32 kHz.

Hardware

The software runs on a standard PC/AT 386 microcomputer with 4 megabytes minimum memory, a 100- or 300-megabyte hard disk and a mouse. A backup device (3M-type cassette for example) is advised as well as an optional digitizer tablet. All Summagraphics-compatible digitizers are supported from size A4 to A0.

The real-time unit is connected to the PC by a bus-to-bus interface (PC/AT—Multibus I). It is made up of three boards (UPICTRL, UPIPROC, and UPISON), which are seen by the PC as memory extensions.

UPICTRL is based on a Fujitsu DSP. It stores the graphic data that represent the pages of music to be played. After preprocessing at a time scale of a few milliseconds, the DSP sends the data to UPIPROC which computes the samples and sends them to UPISON for conversion. This architecture has been designed in order to uncouple the human interface (PC) from the sound-production device (real-time unit).

The memory of the real-time unit (4 megabytes DRAM and 1 megabyte fast SRAM) is large enough to hold as much as four pages of 4000 arcs per page and their associated parameters (64 waveforms, 128 envelopes, and so on) so that it is not necessary for the PC to download any data during sound production. The bandwidth of the link "PC to real-time unit" can be low (typically lower than 1 Megabyte per second). Furthermore, as all the memories are dual-port memories, the PC will be able to modify any parameter of the sound during the computation of the samples. The flexibility of the real-time controls relies on these features.

For example, the timbre of all the arcs that use a certain waveform can be modified and heard in real time when redrawing this waveform: the PC writes the new values into the waveform memory directly, while the same memory is being read for sound computation.

This flexibility, associated with the number of parameters on which it applies, allows composers to create very complex and unpredictable sounds.

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